06 Statistics

1. STATISTICS:

Statistics is a branch of mathematics that deals with the study of collecting, analysing, interpreting, presenting, and organizing data in a particular manner. Statistics is defined as the process of collection of data, classifying data, representing the data for easy interpretation, and further analysis of data. Statistics is also referred to as arriving at conclusions from the sample data that is collected using surveys or experiments.

1.1 Mathematical Statistics:

Statistics is the process of collecting data, evaluating data, and summarizing it into a mathematical form. Initially, statistics were related to the science of the state where it was used in the collection and analysis of facts and data about a country such as its economy, population etc. Mathematical statistics applies mathematical techniques like linear algebra, differential equations, mathematical analysis and theories of probability.

There are two methods of analysing data in mathematical statistics that are used on a large scale:

* Descriptive Statistics
* Inferential Statistics

Rankers Academy JEE

1.2 Descriptive Statistics:

The descriptive method of statistics is used to describe the data collected and summarize the data and its properties using the measures of central tendencies and the measures of dispersion.

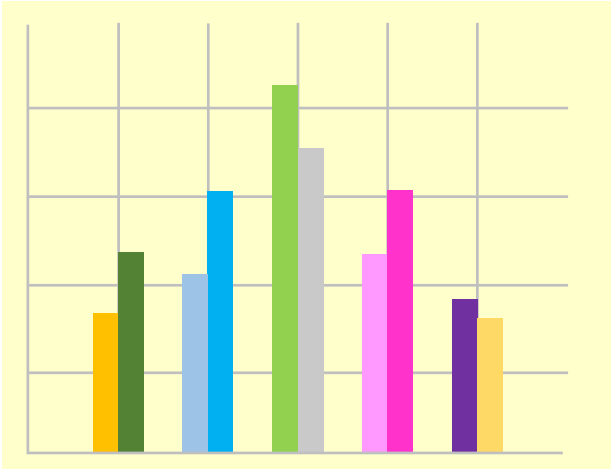
1.3 Inferential Statistics:

This method of statistics is used to draw conclusions from the data. Inferential statistics requires statistical tests performed on samples, and it draws conclusions by identifying the differences between the 2 groups. Tests calculate the p-value that is compared with the probability that the results are due to chance. This alpha (α) which is most often 0.05; therefore, a P-value less than 0.05 is typically considered statistically significant.

1.4 Data Representation in Statistics:

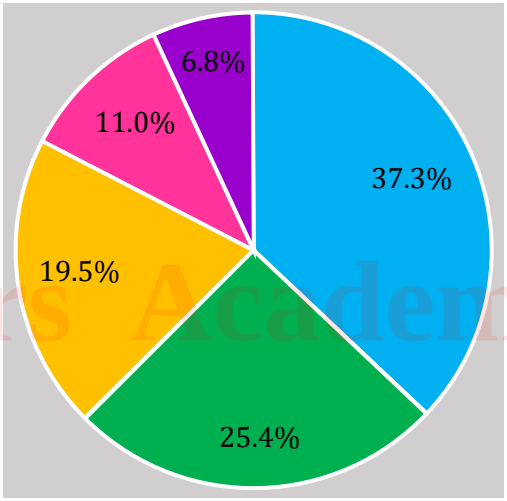
The collection of observations and facts is known a data. These observations and facts can be in the form of numbers, measurements or statements. There are two different kinds of data i.e. Qualitative data and quantitative data. Qualitative data is when the data is descriptive or categorical and quantitative data is when the data is numerical information. Once we know the data collection methods, we aim at representing the collected data in different forms of graphs such as a bar graph, line graph, pie chart, stem and leaf plots, scatter plot, and so on. Before the analysis of data, the outliers are removed that are due to the invariability in the measurements of data. Let us look at different kinds of data representation in statistics.

For More Join: @IITJEE_Advanced



Bar Graph

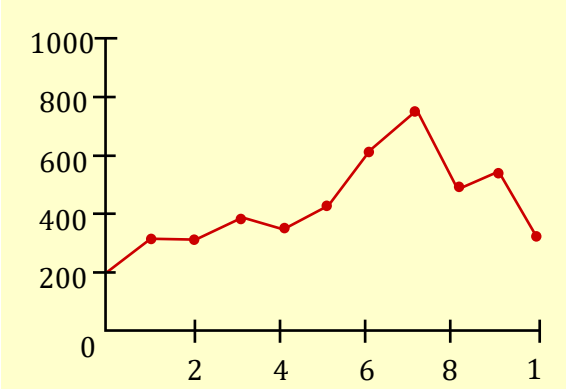
A group of data represented with rectangular bars with lengths proportional to the values is a bar graph. The bars can either be vertically or horizontally plotted.



Rankers Academy JEE

Pie Chart

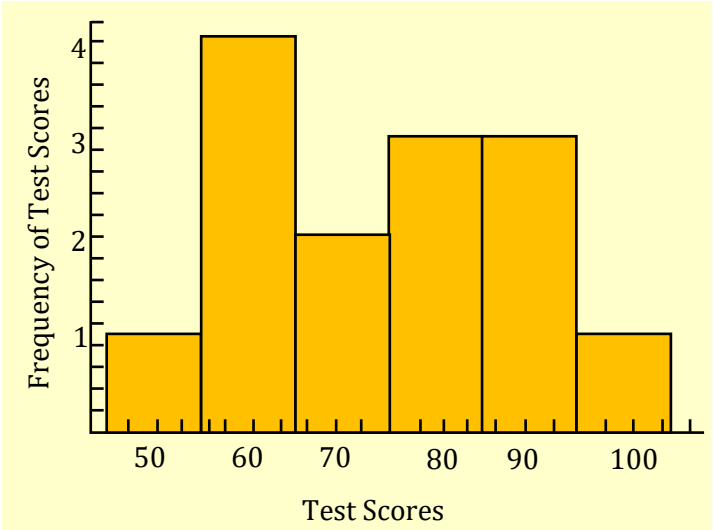
The pie chart is a type of graph in which a circle is divided into sectors where each sector represents a proportion of the whole.



Line graph

The line graph represents the data in a form of series that is connected with a straight line. These series are called markers.

For More Join: @IITJEE_Advanced



Histogram

The histogram is a type of graph where the diagram consists of rectangles, the area is proportional to the frequency of a variable and the width is equal to the class interval. Here is an example of a histogram.

Height	Frequency
139	1
145	2
150	2
136	1
152	1
144	1
138	1

Rankers Academy JEE

- **Frequency Distribution**

  The frequency distribution table in statistics showcases the data in ascending order along with their corresponding frequencies. The frequency of the data is often represented by f.

- **Collection of data**

  The word data means a set of given facts in numerical figures.

- **Fundamental characteristics of data**

  (i) Numerical facts alone form data. Qualitative characteristics, like honesty, poverty, etc., which cannot be measured numerically do not form data.

  (ii) Data are aggregate of facts. A single observation does not form data.

  (iii) Data collected for a definite purpose may not be suitable for another purpose.

  Types of Data

  (i) Primary Data : The data collected by the investigator himself with a definite plan in mind are known as primary data.

  (ii) Secondary Data : The data collected by someone, other than the investigator, are known as secondary data.

For More Join: @IITJEE_Advanced

2. PRESENTATION OF DATA:

2.1 Raw or Ungrouped Data:

The data obtained in original form are called raw data or ungrouped data.

Ex. The marks obtained by 25 students in a class in a certain examination are given below : 25, 8, 37, 16, 45, 40, 29, 12, 42, 40, 25, 14, 16, 16, 20, 10, 36, 33, 24, 25, 35, 11, 30, 45,48.

This is the raw data.

Array : An arrangement of raw data in ascending or descending order of magnitude is called an array.

Arranging the marks of 25 students in ascending order, we get the following array:

8, 10, 11, 12, 14, 16, 16, 16, 20, 24, 25, 25, 25, 29, 30, 33, 35, 36, 37, 40, 40, 42, 45, 45, 48.

2.2 Types of Grouped Frequency Distribution:

2.2.1. Exclusive form (or continuous interval form) :

A frequency distribution in which the upper limit of each class is excluded and lower limit is included, is called an exclusive form.

Ex. Suppose the marks obtained by some students in an examination are given.

We may consider the classes 0 – 10, 10 – 20 etc. In class 0 – 10, we include 0 and exclude 10. In class 10 – 20, we include 10 and exclude 20.

2.2.2 Inclusive Form (or Discontinuous Interval Form) : A frequency distribution in which each upper limit as well as lower limit is included, is called an inclusive form. Thus, we have classes of the form 0 – 10, 11 – 20, 21 – 30 etc. In 0 – 10, both 0 and 10 are included.

Rankers Academy JEE

3. GRAPHICAL REPRESENTATION OF STATISTICAL DATA:

The tabular representation of data is an ideal way of presenting them in a systematic manner. When these numerical figures are represented pictorially or graphically, they become more noticeable and easily intelligible, leaving a more lasting effect on the mind of the observer. With the help of these pictures or graphs, data can be compared easily.

There are various types of graphs. In this chapter, we shall be dealing with the following graphs:

1. Bar graphs 2. Histogram 3. Frequency polygon

3.1 Bar Graph (or Column Graph or Bar Chart):

A bar graph is a pictorial representation of numerical data in the form of rectangles (or bars) of equal width and varying heights.

These rectangles are drawn either vertically or horizontally.

The height of a bar represents the frequency of the corresponding observation. The gap between two bars is kept the same.

For More Join: @IITJEE_Advanced

3.2 Histogram:

A histogram is a graphical representation of a frequency distribution in an exclusive form in the form of rectangles with class intervals as bases and the corresponding frequencies as heights, there being no gap between any two successive rectangles.

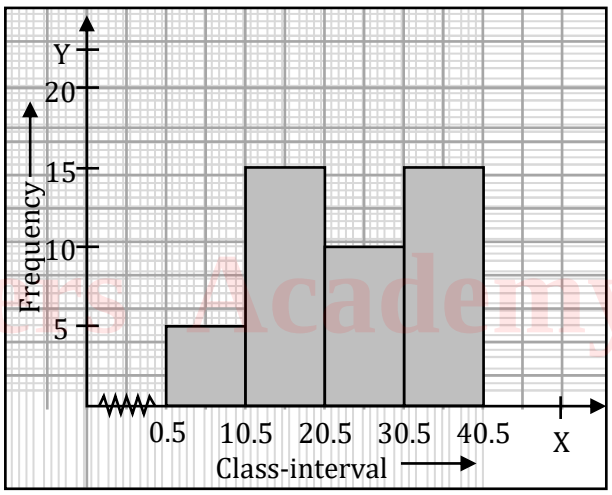
Method of drawing a histogram

Step 1 : If the given frequency distribution is in inclusive form, convert it into an exclusive form.

Step 2 : Taking suitable scales, mark the class-intervals along x-axis and frequencies along y-axis. Note that the scales chosen for both the axes need not be the same.

Step 3 : Construct rectangles with class-intervals as bases and the corresponding frequencies as heights.

Note : Since the scale on x-axis starts at 0.5, a kink is shown near the origin.



Rankers Academy JEE

A random survey of the number of children of various age groups playing in a park was found as follows.

Age (in years)	1–2	2–3	3–5	5–7	7–10	10–15	15–17
Number of Children	5	3	6	12	9	10	4

Draw a histogram to represent the data above.

Explanation

In the given frequency distribution, we see that the class size are different. Hence, we calculate the adjusted frequency for each class by using the formula.

$$\text{Adjusted frequency for a class} = \frac{\text{Minimum class size}}{\text{Class size of the class}} \times \text{frequency of the class}$$

In the given distribution, the minimum class size is 1.

The adjusted frequencies are computed in the following table.

For More Join: @IITJEE_Advanced

Age (in years)	Frequency	Class size	Adjusted Frequency
1-2	5	1	$\frac{1}{1}\times 5=5$
2-3	3	1	$\frac{1}{1}\times 3=3$
3-5	6	2	$\frac{1}{2}\times 6=3$
5-7	12	2	$\frac{1}{2}\times 12=6$
7-10	9	3	$\frac{1}{3}\times 9=3$
10-15	10	5	$\frac{1}{5}\times 10=2$
15-17	4	2	$\frac{1}{2}\times 4=2$

3.3 Frequency Polygon:

Let x_1, x_2, ..., x_n be the class marks (i.e., mid points) of the given frequency distribution and let f_1, f_2, ... f_n be the corresponding frequencies. We plot the points (x_1, f_1), (x_2, f_2),..., (x_n, f_n) on a graph paper and join these points by line segments. We complete the diagram in the form of a polygon by taking two more classes (called imagined classes), one at the beginning and the other at the end, each with frequency zero. This polygon is known as the frequency polygon of the given frequency distribution.

Rankers Academy JEE

Illustration 1

The following table shows the number of diabetic patients at various age groups.

Age (in years)	10-20	20-30	30-40	40-50	50-60	60-70
Number of diabetic patients	3	6	14	8	5	2

Represent the above data by a frequency polygon.

Solution:

Take two more intervals one at the beginning and the other at the end, each with frequency 0.

Thus, we have class-intervals 0 – 10, 10 – 20, 20 – 30, 30 – 40, 40 – 50, 50 – 60, 60 – 70 and 70 – 80 with corresponding frequencies as 0, 3, 6, 14, 8, 5, 2, 0 respectively. Thus, we have:

Class mark	5	15	25	35	45	55	65	75
Frequency	0	3	6	14	8	5	2	0

Now, plot the points (5, 0), (15, 3), (25, 6), (35, 14), (45, 8), (55, 5), (65, 2), (75, 0) on a graph paper and join them successively to get the required graph.

For More Join: @IITJEE_Advanced

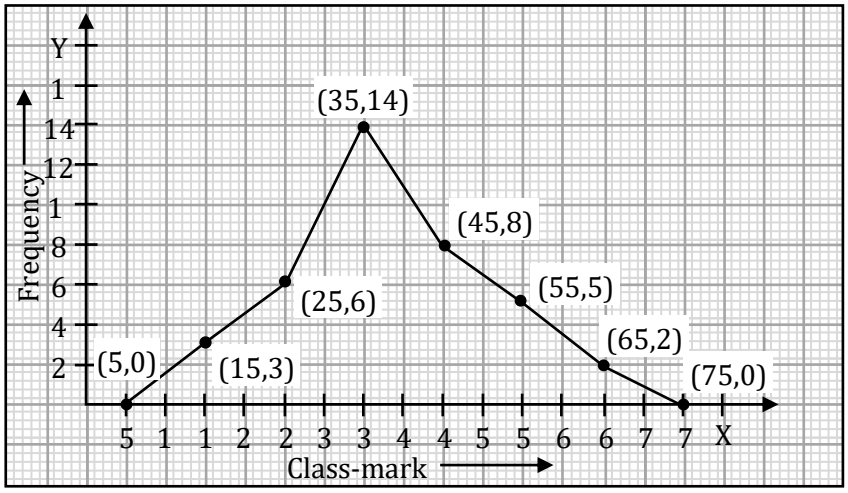


Illustration 2

A survey conducted by an organisation for the cause of illness and death among the women between the ages 15 – 44 (in years) worldwide, found the following figures (in %):

S.No.	Causes	Female fatality rate (%)
1.	Reproductive health conditions	31.8
2.	Neuropsychiatric conditions	25.4
3.	Injuries	12.4
4.	Cardiovascular conditions	4.3
5.	Respiratory conditions	4.1
6.	Other causes	22

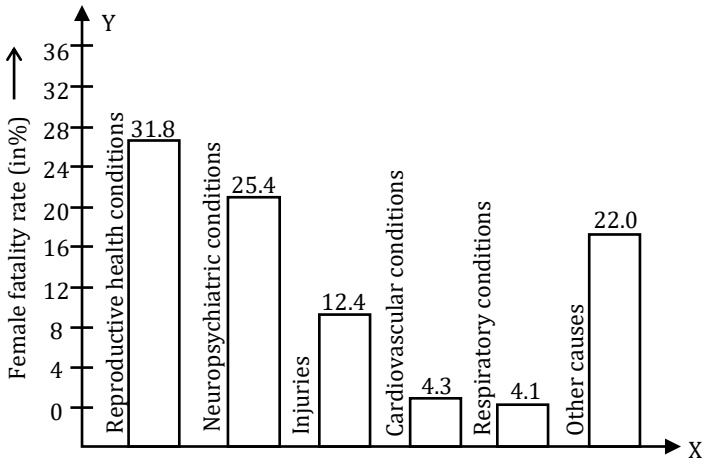
(i) Represent the information given above graphically.

(ii) Which condition is the major cause of women's ill health and death worldwide?

Bankers Academy JEE

Solution:

(i)



(ii) Reproductive health conditions is the major cause of women's ill health and death worldwide.

MEASURES OF CENTRAL TENDENCY (OR AVERAGE VALUES):

An Average value of a distribution is the value of variable which is representative of the whole distribution. Generally, average values lies in the central part of the distribution and they are also called measures of central tendency.

The following five measures of central tendency can be divided in two groups.

For More Join: @IITJEE_Advanced

(I) Mathematical Average:

(i) Arithmetic Mean or Mean

(ii) Geometric Mean

(iii) Harmonic Mean

(II) Positional Average:

(i) Median

(ii) Mode

ARITHMETIC MEAN (A.M.):

(i) For Ungrouped (or Individual) Distribution:

If $x_1, x_2, ..., x_n$ are n values of variate then their arithmetic mean $(A.M.) = \frac{x_1 + x_2 + ... + x_n}{n}$

$$\boxed{\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}} \text{ or } \boxed{\sum_{i=1}^{n} x_i = n\bar{x}}$$

Illustration 3

In a group of 30 observations mean of first 10 is 12 and last 20 is 9, then find mean of whole distribution.

Solution:

Sum of first 10 observation $= 10 \times 12 = 120$

Sum of last 20 observation $= 20 \times 9 = 180$

Sum of 30 observation $= 120 + 180 = 300$

Mean $= \frac{\sum x_i}{n} = \frac{300}{30} = 10$

Rankers Academy JEE

Illustration 4

Mean of n observation $1^2, 2^2, 3^2, ... n^2$ is $\frac{46n}{11}$. Then value of n is :

(A) 11 (B) 12 (C) 23 (D) 22

Ans. (A)

Solution:

$$\bar{x} = \frac{\sum x_i}{n} \Rightarrow \frac{46n}{11} = \frac{1^2 + 2^2 + ...n^2}{n}$$

$$\Rightarrow \frac{46n}{11} = \frac{n(n+1)(2n+1)}{6n}$$

Solving

$n = 11$

Illustration 5

Find the $A.M.$ of the series $1,2,4,8,16, ..., 2^n$

(A) $\frac{2^{n+1} - 1}{n+1}$ (B) $\frac{2^{n+2} - 1}{n}$ (C) $\frac{2^n - 1}{n+1}$ (D) $\frac{2^n - 1}{n}$

Ans. ()

Solution:

$$\bar{x} = \frac{1 + 2 + 4 ..... 2^n}{n+1} = \frac{2^{n+1} - 1}{2 - 1} \cdot \frac{1}{n+1}$$

For More Join: @IITJEE_Advanced

ARITHMETIC MEAN (A.M.):

For Ungrouped Frequency Distribution:

If $x_1, x_2, \ldots\ldots, x_n$ are n values of variate x_i and their frequencies are $f_1, f_2, \ldots\ldots, f_n$ respectively, then AM of this frequency distribution is -

$$\bar{x} = \frac{f_1x_1 + f_2x_2 + \ldots\ldots + f_nx_n}{(f_1 + f_2 + \ldots\ldots + f_n)}$$

$$\Rightarrow \bar{x} = \frac{\sum_{i=1}^{n} f_ix_i}{\sum_{i=1}^{n} f_i} \quad \text{or} \quad \boxed{\bar{x} = \frac{\sum_{i=1}^{n} f_ix_i}{N}} \quad \text{Where } N = \sum_{i=1}^{n} f_i$$

Note: For AM of grouped frequency distribution, we find the middle values of each group (or class) and changed it into ungrouped frequency distribution, then use above formula.

Illustration 6

Find the A.M. of the following frequency distribution.

x_i	5	8	11	14	17
f_i	4	5	6	10	20

Solution:

Rankers Academy JEE

Here, $N = \Sigma f_i = 4 + 5 + 6 + 10 + 20 = 45$

$\Sigma f_i x_i = (5 \times 4) + (8 \times 5) + (11 \times 6) + (14 \times 10) + (17 \times 20) = 606$

$$\therefore \ \bar{x} = \frac{\Sigma f_i x_i}{N} = \frac{606}{45} = 13.47$$

Illustration 7

Find the mean marks from the following data :

Marks	No. of Students
Below 10	5
Below 20	9
Below 30	18
Below 40	29
Below 50	45
Below 60	60
Below 70	70
Below 80	78
Below 90	83
Below 100	85

For More Join: @IITJEE_Advanced

Solution:

We may prepare the table as given below :

Marks	No. of students	Class Interval	f_i	Class mark (x_i)	$f_i x_i$
Below 10	5	0-10	5	5	25
Below 20	9	10-20	4	15	60
Below 30	18	20-30	9	25	225
Below 40	29	30-40	11	35	385
Below 50	45	40-50	16	45	720
Below 60	60	50-60	15	55	825
Below 70	70	60-70	10	65	650
Below 80	78	70-80	8	75	600
Below 90	83	80-90	5	85	425
Below 100	85	90-100	2	95	190
			$N = 85$		$\sum f_i x_i = 4105$

$\therefore$ Mean, $\bar{x} = \dfrac{\sum f_i x_i}{N} = \dfrac{4105}{85} = 48.29$

Short Method/Assumed Mean Method (By Change of Origin):

In this method, we take deviation (d_i) of each variate (x_i) from any central value (a).

Let $x_i - a = d_i \Rightarrow x_i = a + d_i$

Then $\bar{x} = \dfrac{\sum f_i(a+d_i)}{N} = \dfrac{a\sum f_i}{N} + \dfrac{\sum f_i d_i}{N}$

$\boxed{\bar{x} = a + \dfrac{\sum f_i d_i}{N}}$ where a is assumed mean (any central value of variate x_i)

Rankers Academy JEE

Step Deviation Method (By Change of Origin and Scale):

In this method, if each value of deviation (d_i) is divisible by any common number h (let)

Let, $\dfrac{d_i}{h} = u_i \Rightarrow d_i = hu_i$

Then, $\boxed{\bar{x} = a + \dfrac{\sum f_i u_i}{N} \times h}$

Illustration 8

Find the $A.M$ of the following frequency distribution

Wage (in Rs.)	800	820	860	900	920	980	1000
No. of Worker	7	14	19	24	20	11	5

Solution:

Let $a = 900, h = 20$

x_i	f_i	$d_i = x_i$	$u_i = d_i/h$	$f_i u_i$
800	7	–100	–5	–35
820	14	–80	–4	–56
860	19	–40	–2	–38
900	24	0	0	0
920	20	20	1	20
980	11	80	4	44
1000	5	100	5	25
	$N = 100$			$\sum f_i u_i = -40$

For More Join: @IITJEE_Advanced

$\because \ \bar{x} = a + \frac{\sum f_i u_i}{N} \times h$

$= 900 + \frac{(-40)}{100} \times 20$

$= 900 - 8$

$= 892$

(v) Weighted Mean (W.M.):

If $x_1, x_2, \ldots\ldots, x_n$ are n values of variate x_i and their weightage are $w_1, w_2, \ldots\ldots, w_n$ respectively, then their weighted mean is -

$$W.M. = \frac{w_1x_1 + w_2x_2 + \ldots\ldots + w_nx_n}{w_1 + w_2 + \ldots\ldots + w_n} = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}$$

Illustration 9

Find the weighted mean of first n natural numbers when their weights are equal to their squares respectively

Solution:

$$\text{Weighted Mean} = \frac{1.1^2 + 2.2^2 + \ldots. + n.n^2}{1^2 + 2^2 + \ldots. + n^2} = \frac{1^3 + 2^3 + \ldots.. + n^3}{1^2 + 2^2 + \ldots. + n^2} = \frac{[n(n+1)/2]^2}{[n(n+1)(2n+1)/6]} = \frac{3n(n+1)}{2(2n+1)}$$

(vi) Combined Mean (C.M.):

Rankers Academy JEE

If $\bar{x}_1, \bar{x}_2, \ldots\ldots, \bar{x}_k$ are mean of observations of k series which have $n_1, n_2, \ldots\ldots, n_k$ terms respectively, then their combined mean is-

$$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + \ldots\ldots + n_k\bar{x}_k}{n_1 + n_2 + \ldots\ldots + n_k}$$

If $k = 2$, then $\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$

Illustration 10

The mean income of a group of persons is Rs. 400 and another group of persons is Rs. 480. If the mean income of all the persons of these two groups is Rs. 430, then find the ratio of the number of persons in the groups.

Solution:

Here $\bar{x}_1 = 400, \ \bar{x}_2 = 480, \ \bar{x} = 430$

$\because \ \bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2} \Rightarrow 430 = \frac{400n_1 + 480n_2}{n_1 + n_2}$

$\Rightarrow \frac{n_1}{n_2} = \frac{5}{3}$

Illustration 11

Average salary of male employee in a firm was Rs. 5200 and of female employee was Rs. 4200. The mean salary of all employee was Rs. 5000, then percentage of male employee in a firm is:

For More Join: @IITJEE_Advanced

Solution:

$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$

$5000 = \frac{5200n_1 + 4200n_2}{n_1 + n_2} \Rightarrow n_1 = 4n_2$

Percentage of male = $\frac{n_1}{n_1 + n_2} \times 100$

$= \frac{4n_2}{5n_2} \times 100 = 80\%$

Properties of AM:

(1) The sum of deviation of variate from their *AM* is always zero i.e. $\sum(x_i - \bar{x}) = 0$, $\sum f_i(x_i - \bar{x}) = 0$

(2) The sum of square of deviation of variate from their *AM* is minimum i.e. $\sum(x_i - \bar{x})^2 \le \sum(x_i - a)^2$

(where a is any central value)

(3) If the *AM* of the observations $x_1, x_2, \ldots\ldots, x_n$ is $\bar{x}$, then

- *AM* of $x_1 \pm \lambda, x_2 \pm \lambda, \ldots\ldots, x_n \pm \lambda$ is $\bar{x} \pm \lambda$
- *AM* of $\lambda x_1, \lambda x_2, \ldots\ldots, \lambda x_n$ is $\lambda\bar{x}$
- *AM* of $ax_1 \pm b, ax_2 \pm b, \ldots\ldots, ax_n \pm b$ is a $\bar{x} \pm b$ (where λ, a, b are constant)

(4) *AM* is not affected by any change in assumed mean.

Illustration 12

Mean of 16 observations is 16. If one observation of value 16 is deleted and 3 observations 3, 4, 5 are added to this set, then find mean of new set of observations.

Solution:

Sum of 16 observations $= 16 \times 16 = 256$

New sum = $256 - 16 + (3 + 4 + 5) = 252$

New mean = $\frac{\sum x_i}{n} = \frac{252}{18} = 14$

Illustration 13

Mean age of 25 teacher in a school is 40 years. A teacher is retired at age of 60 years and a new teacher is appointed in his place. If new mean age of teacher in school is 39 years. Then age of newly appointed teacher is

(A) 35 (B) 30 (C) 40 (D) 25

Solution:

Sum of age of 25 teachers $= 25 \times 40 = 1000$ years.

Let age of newly appointed teacher is x years

New sum = $1000 - 60 + x$

New mean = $\frac{1000 - 60 + x}{25}$

$39 = \frac{940 + x}{25}$

$\Rightarrow x = 35$ years

For More Join: @IITJEE_Advanced

4.2 Graphical Representation of Cumulative Frequency Distribution:

- **Cumulative frequency polygon curve (Ogive)**

  Cumulative frequency is of two types and corresponding to these, the ogive is also of two types.

  - **Less than series**
  - **More than series**

- **Less than series :** To construct a cumulative frequency polygon and an ogive, we follow these steps :

  Step-1 : Mark the upper class limit along x-axis and the corresponding cumulative frequencies along y-axis.

  Step-2 : Plot these points successively by line segments. We get a polygon, called cumulative frequency polygon.

  Step-3 : Plot these points successively by smooth curves, we get a curve called cumulative frequency curve or an ogive.

- **More than series :** To construct a cumulative frequency polygon and an ogive, we follow these steps:

  Step-1 : Mark the lower class limits along x-axis and the corresponding cumulative frequencies along y-axis.

  Step-2 : Plot these points successively by line segments, we get a polygon, called cumulative frequency polygon.

  Step-3 : Plot these points successively by smooth curves, we get a curve, called cumulative frequency curve or an ogive.

  Application of an ogive

  Ogive can be used to find the median of a frequency distribution. To find the median, we follow these steps.

  Method-I

  Step-1 : Draw anyone of the two types of frequency curves on the graph paper.

  Step-2 : Compute $\frac{N}{2}(N=\Sigma f_i)$ and mark the corresponding points on the y-axis.

  Step-3 : Draw a line parallel to x-axis from the point marked in step 2, cutting the cumulative frequency curve at a point P.

  Step-4 : Draw perpendicular PM from P on the x-axis. The x- coordinate of point M gives the median.

  Method-II

  Step-1 : Draw less than type and more than type cumulative frequency curves on the graph paper.

  Step-2 : Mark the point of intersection (P) of the two curves drawn in step 1.

  Step-3 : Draw perpendicular PM from P on the x-axis. The x- coordinate of point M gives the median.

Rankers Academy JEE

Illustration 14

The following distribution gives the daily income of 50 workers of a factory.

Daily income (in Rs.)	100-120	120-140	140-160	160-180	180-200
No. of workers	12	14	8	6	10

Convert the distribution above to a less than type cumulative frequency distribution and draw its ogive.

Solution:

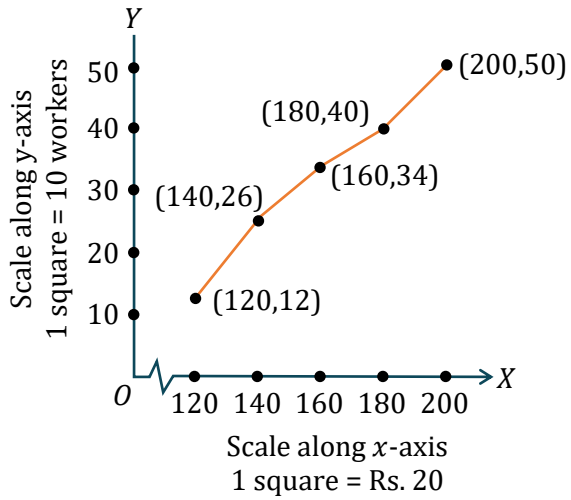
From the given table, we prepare a less than type cumulative frequency distribution table, as given below:

Income less than (in Rs.)	120	140	160	180	200
Cumulative frequency	12	26	34	40	50

Now, plot the points (120, 12), (140, 26), (160, 34), (180, 40) and (200, 50).

For More Join: @IITJEE_Advanced

Join these points by a freehand curve to get an ogive of 'less than' type.



MEDIAN:

Median is the middle value of a series when the values of the series are arranged in ascending or descending order therefore median divides an arranged series into two equal parts.

(i) For Ungrouped Distribution:

Let n be the number of terms of series, first we arrange the series in the ascending or descending order, then in this arranged series:

$$\text{Median} = \begin{cases} \left(\frac{n+1}{2}\right)^{\text{th}} \text{ term} & (\text{when } n \text{ is odd}) \\ \dfrac{\left(\frac{n}{2}\right)^{\text{th}} \text{ term} + \left(\frac{n}{2}+1\right)^{\text{th}} \text{ term}}{2} & (\text{when } n \text{ is even}) \end{cases}$$

Rankers Academy JEE

(ii) For Ungrouped Frequency Distribution:

(a) First, we arrange the value of variate x_i, in ascending order with their frequency.

(b) Prepare cumulative frequency $(c.f.)$ column and find the value of N.

$$\text{Median} = \begin{cases} \left(\frac{N+1}{2}\right)^{\text{th}} \text{ term} & (\text{when } N \text{ is odd}) \\ \dfrac{\left(\frac{N}{2}\right)^{\text{th}} \text{ term} + \left(\frac{N}{2}+1\right)^{\text{th}} \text{ term}}{2} & (\text{when } N \text{ is even}) \end{cases}$$

Illustration 15

Find the median of following frequency distribution.

x_i	42	52	62	72	47	57	67
f_i	3	6	11	9	8	8	5

Solution:

x_1	f_i	$C.f$
42	3	3
47	8	11
52	6	17
57	8	25
62	11	36
67	5	41
72	9	50

For More Join: @IITJEE_Advanced

Here $N = 50$ (even)

$$\text{Median} = \frac{\left(\frac{N}{2}\right)^{th} \text{ term} + \left(\frac{N}{2}+1\right)^{th} \text{ term}}{2}$$

$$= \frac{25^{th} \text{ term} + 26^{th} \text{ term}}{2}$$

$$= \frac{57+62}{2} = \frac{119}{2} = 59.5$$

(iii) For Grouped Frequency Distribution:

(a) Prepare the cumulative frequency ($c.f.$) column and find the value of $\frac{N}{2}$

(b) Find the class which contain the value of $c.f.$ equal or just greater to $\frac{N}{2}$. This is median class.

$$\text{Median} = \ell + \frac{\left(\frac{N}{2} - F\right)}{f} \times h$$

Where ℓ — lower limit of median class.

F — $c.f.$ of the class preceding median class.

f — frequency of median class.

h — Class interval of median class.

Illustration 16

Find the median of following frequency distribution.

Class	0–5	5–10	10–15	15–20	20–25
f_i	6	10	8	9	7

Solution:

Class	f_i	$C.f$
0–5	6	6
5–10	10	16
10–15	8	24
15–20	9	33
20–25	7	40

Here $\frac{N}{2} = \frac{40}{2} = 20$

Median Class is 10– 15

$\therefore$ Median $= 10 + \frac{(20-16)}{8} \times 5$

$= 10 + 2.5 = 12.5$

Illustration 17

Find the median of the following frequency distribution.

Class	1–10	11–20	21–30	31–40	41–50
f_i	5	6	8	7	4

Rankers Academy JEE
For More Join: @IITJEE_Advanced

Solution:

Class	f_i	$C.f$
0.5–10.5	5	5
10.5–20.5	6	11
20.5–30.5	8	19
30.5–40.5	7	26
40.5–50.5	4	30

Here $\frac{N}{2} = \frac{30}{2} = 15$

Median class is 20.5 – 30.5

$\therefore$ Median = $20.5 + \frac{(15-11)}{8} \times 10$

$= 20.5 + 5 = 25.5$

Illustration 18

Set of 100 observation is divided into equal class interval whose median is 25. If median class is 20-30 and 45 observation is below from 20 then $'f'$ is:

(A) 10 (B) 20 (C) 30 (D) 15

Ans. (A)

Solution:

$$\text{Median} = \ell + \frac{\left(\frac{N}{2} - F\right)}{f} \times h$$

Rankers Academy JEE

$$25 = 20 + \frac{\frac{100}{2} - 45}{f} \times 10$$

On solving, we get f =10

MODE:

Mode is the value of variate which have maximum frequency.

(i) For ungrouped distribution

The value of variate which is repeated maximum number of times.

(ii) For ungrouped frequency distribution.

The value of variate which have maximum frequency.

(iii) For grouped frequency distribution.

First, we find the class which have maximum frequency. This is modal class, then

$$\text{Mode} = \ell + \frac{(f_0 - f_1)}{(2f_0 - f_1 - f_2)} \times h$$

Where ℓ — lower limit of modal class.

f_0 — frequency of modal class.

f_1 — frequency of the class preceding modal class.

f_2 — frequency of the class succeeding modal class.

h — class interval of modal class.

For More Join: @IITJEE_Advanced

Illustration 19

Find the mode of the following frequency distribution.

x_i	4	5	6	7	8	9
f_i	3	4	5	8	7	6

Solution:

Here the variate 7 have maximum frequency.

$\therefore$ Mode = 7

Illustration 20

Find the mode of the following frequency distribution

Class	0–10	10–20	20–30	30–40	40–50	50–60	60–70	70–80
f_i	2	18	30	45	35	20	6	3

Solution:

Here the class 30–40 has maximum freq. so this is the modal class

$\ell = 30, f_0 = 45, f_1 = 30, f_2 = 35, h = 10$

$$\therefore \text{ Mode} = \ell + \frac{f_0 - f_1}{2f_0 - f_1 - f_2} \times h = 30 + \frac{45-30}{2\times 45 - 30 - 35} \times 10 = 36$$

RELATION BETWEEN MEAN, MEDIAN AND MODE:

In a moderately asymmetric distribution following relation between mean, median and mode of a distribution. It is known as **Empirical formula.**

Mode = 3 Median – 2 Mean

Leaders Academy JEE

Illustration 21

If difference between mean and mode is 3, the difference between mean and median is

(A) 3 (B) 1 (C) 4 (D) 2

Ans. (B)

Solution:

Mode + 2 Mean = 3 Median

(Mean –3) + 2 Mean = 3 Median

3(Mean – Median) = 3

$\Rightarrow$ Mean – Median = 1

MEASURES OF DISPERSION:

What is Dispersion?

Dispersion refers to how "spread out" a group of scores is. To see what we mean by spread out, consider table. These table represent the scores on two quizzes. The mean score for each quiz is 7.0. Despite the equality of means, you can see that the distributions are quite different. Specifically, the scores on Quiz 1 are more densely packed and those on Quiz 2 are more spread out. The differences among students were much greater on Quiz 2 than on Quiz 1.

Quiz 1	2	6	5	4	3		
Quiz 2	2	4	3	3	2	3	3

For More Join: @IITJEE_Advanced

The terms dispersion, spread, and variability are synonyms, and refer to how spread out a distribution is. Just as in the section on central tendency where we discussed measures of the centre of a distribution of scores, in this chapter we will discuss measures of the dispersion of a distribution. There are four frequently used measures of dispersion: range, interquartile range, variance and standard deviation.

The dispersion of a distribution is the measure of deviation of variates from their average value.

Generally, the following measures of dispersion are used.

(1) Range

(2) Mean Deviation

(3) Variance and Standard Deviation

RANGE:

The range of a distribution is the difference of highest value (H) and least value (L) of the distribution.

The Range of the grouped distribution is the difference of upper limit of maximum class (H) and lower limit of minimum class (L). Therefore

Range = $H - L$

Coefficient of Range = $\frac{\text{Difference of extreme values}}{\text{Sum of extreme values}} = \frac{H-L}{H+L}$

Illustration 22

Range of data 7, 8, 2, 1, 3, 13, 18 is

(A) 10 (B) 15 (C) 17 (D) 11

Ans. (C)

Solution:

Range of data = difference of extreme values $= 18 - 1 = 17$

Illustration 23

Coefficient of range 5, 2, 3, 4, 6, 8, 10 is

(A) $\frac{2}{3}$ (B) $\frac{1}{3}$ (C) $\frac{3}{5}$ (D) $\frac{1}{2}$

Ans. (A)

Solution:

Coefficient of range = $\frac{\text{Difference of extreme values}}{\text{Sum of extreme values}} = \frac{10-2}{10+2} = \frac{2}{3}$

MEAN DEVIATION (M.D.):

The mean of absolute value of deviation of each variate from any central value is called mean deviation about that central value.

Let A be any central value (Mean, Median, Mode) of a dist. then M.D. about A are defined as

$M.D. = \frac{\sum_{i=1}^{n} |x_i - A|}{n}$ (For ungrouped dist.)

$M.D. = \frac{\sum_{i=1}^{n} f_i |x_i - A|}{N}$ (For frequency dist.)

Rankers Academy JEE
For More Join: @IITJEE_Advanced

Illustration 24

Find the mean deviation of number 3, 4, 5, 6, 7

Solution:

Here $n = 5$, $\bar{x} = 5$

$\therefore$ Mean deviation $= \dfrac{\Sigma|x_i - \bar{x}|}{n}$

$= \dfrac{1}{5}[|3-5| + |4-5| + |5-5| + |6-5| + |7-5|]$

$= \dfrac{1}{5}[2 + 1 + 0 + 1 + 2] = \dfrac{6}{5} = 1.2$

Illustration 25

Find the $M.D.$ of the following numbers about their median 34, 38, 42, 44, 46, 48, 54, 55, 63, 70

Solution:

Here $n = 10$ (even)

Median $(M) = \dfrac{\left(\frac{n}{2}\right)^{th}\text{ term} + \left(\frac{n}{2}+1\right)^{th}\text{ term}}{2} = \dfrac{5^{th}\text{ term} + 6^{th}\text{ term}}{2} = \dfrac{46+48}{2} = 47$

$\because \sum|x_i - M| = 13 + 9 + 5 + 3 + 1 + 1 + 7 + 8 + 16 + 23 = 86$

$\therefore M.D. = \dfrac{\Sigma|x_i - M|}{n} = \dfrac{86}{10} = 8.6$

Rankers Academy JEE

Illustration 26

Find the mean deviation about mean from the following data

x_i	3	9	17	23	27
f_i	8	10	12	9	5

Solution:

x_i	f_i	$f_i x_i$	$\lvert x_i - \bar{x}\rvert$	$f_i\lvert x_i - \bar{x}\rvert$
3	8	24	12	96
9	10	90	6	60
17	12	204	2	24
23	9	207	8	72
27	5	135	12	60
	$N = 44$	$\sum f_i x_i = 660$		$\sum f_i\lvert x_i - \bar{x}\rvert = 312$

Mean $(\bar{x}) = \dfrac{\Sigma f_i x_i}{N} = \dfrac{660}{44} = 15$

Mean deviation $= \dfrac{\Sigma f_i|x_i - \bar{x}|}{N} = \dfrac{312}{44} = 7.09$

Illustration 27

Find the $M.D.$ of the following frequency distribution about median.

Class	0–10	10–20	20–30	30–40	40–50
f_i	2	8	10	4	6

For More Join: @IITJEE_Advanced

Solution:

Class	f_i	$c.f$	x_i	$f_i\|x_i - M\|$
0–10	2	2	5	2.2=40
10–20	8	10	15	8.10=80
20–30	10	20	25	10.0=0
30–40	4	24	35	4.10=40
40–50	6	30	45	6.20=120
	$N = 30$			$\sum f_i\|x_i - M\| = 280$

Here $\frac{N}{2} = 15$, Median class is 20– 30

$\therefore$ Median $= \ell + \frac{\left(\frac{N}{2} - F\right)}{f} \times h = 20 + \frac{(15-10)}{10} \times 10 = 25$

$\therefore$ $M.D.$ about Median $= \frac{\sum f_i \,|\, x_i - M \,|}{N} = \frac{280}{30} = 9.33$

Illustration 28

If mean deviation of number $1, 1 + d, 1 + 2d, \ldots\ldots 1 + 50\, d$ from their mean is 325, then value of $'d'$ is :

Solution:

$x_i \rightarrow 1, 1 + d, 1 + 2d, \ldots\ldots 1 + 50\, d$

Mean $= 1 + 25\, d$

$M.D. = \frac{\sum f_i \,|\, x_i - (1+25d) \,|}{n}$

$325 = \frac{25d + 24d + \ldots\ldots + d + 0 + d + \ldots . 25d}{51}$

$325 = \frac{2d[1 + 2 + \ldots\ldots + 25]}{51}$

$\frac{325 \times 51}{2} = d \cdot \frac{25(26)}{2}$

$d = \frac{51}{2}$

$d = 25.5$

Illustration 29

If mean deviation of number $a, 2a, 3a, \ldots\ldots, 50a$ from median is 50, then $|a|$ equals:

Solution:

$x_i \rightarrow a, 2a, 3a, \ldots\ldots 50a$

Median $= 25.5a$

M.D. $= \frac{\sum f_i \,|\, x_i - 25.5a \,|}{n}$

$50 = \frac{24.5a + 23.5a + \ldots\ldots + 0.5a + \ldots\ldots 24.5a}{50}$

$2500 = \frac{2a}{10}(5 + 15 + 25 + \ldots\ldots + 245)$

$2500 = \frac{2a}{10} \times \frac{25}{2}(5 + 245)$

$a = 4$

$\therefore$ $|a| = 4$

Rankers Academy JEE

For More Join: @IITJEE_Advanced

VARIANCE AND STANDARD DEVIATION:

Variance:

The Mean of square of deviation of each variate from their mean is called variance, it is denoted by σ^2.

Standard Deviation (S.D.):

The positive square root of the variance are called standard deviation. It is denoted by σ.

i.e., $\boxed{\text{S.D.} = +\sqrt{\text{variance}}}$

Formulae for Variance:

(i) For ungrouped distribution

$$\sigma^2 = \frac{\sum(x_i - \bar{x})^2}{n}$$

$$\sigma^2 = \frac{\sum x_i^2}{n} - \bar{x}^2 = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2$$

$$\sigma^2 = \frac{\sum d_i^2}{n} - \left(\frac{\sum d_i}{n}\right)^2 \qquad \text{Where } d_i = x_i - a$$

$$\sigma^2 = h^2\left[\frac{\sum u_i^2}{n} - \left(\frac{\sum u_i}{n}\right)^2\right] \qquad \text{Where } u_i = \frac{d_i}{h}$$

Illustration 30

Find variance of first 'n' even natural number is:

Solution:

$x_i \to 2,4,6,\ldots\ldots,2n$

$$\bar{x} = \frac{\sum x_i}{n} = \frac{n(n+1)}{n} = n+1$$

Rankers Academy JEE

$$\text{Variance } (x_i) = \frac{\sum x_i^2}{n} - (\bar{x})^2$$

$$= \frac{2^2 + 4^2 + \ldots.(2n)^2}{n} - (n+1)^2$$

$$= 4 \times \frac{n(n+1)(2n+1)}{6n} - (n+1)^2$$

$$= \frac{4}{6}(n+1)(2n+1) - (n+1)^2 = \frac{n^2-1}{3}$$

Illustration 31

The sum of 100 observation and sum of there square are 400 and 2475 respectively. Later on three observation 3, 4, 5 are found incorrect. If these incorrect data is omitted then variance of remaining observation is :

(A) 8.00 (B) 8.25 (C) 9.00 (D) 9.25

Ans. (C)

Solution:

New $\sum x_i = 400 - (3 + 4 + 5) = 400 - 12 = 388$

New $\sum x_i^2 = 2475 - (3^2 + 4^2 + 5^2) = 2475 - 50 = 2425$

$$\text{So, var } (x_i) = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2$$

$$= \frac{2425}{97} - \left(\frac{388}{97}\right)^2 = 25 - 16 = 9$$

For More Join: @IITJEE_Advanced

Illustration 32

If mean and variance of data $a, b, 8, 5, 10$ is 6 and 6.8 respectively. Then value of and b is :

Solution:

Mean = 6

$$\frac{a+b+8+5+10}{5}=6$$

$a + b = 7$ ...(i)

and var $(x_i)=6.8$

$$\frac{a^2+b^2+8^2+5^2+10^2}{5}-(6)^2=6.8$$

$a^2 + b^2 = 25$ ...(ii)

from (i) and (ii)

$$\begin{cases}a=3,b=4\\a=4,b=3\end{cases}$$

Illustration 33

If $S.D.$ of number $2, 3, a$ and 11 is 3.5, then which of them is true:

(A) $3a^2 - 34a + 91 = 0$ (B) $3a^2 - 23a + 44 = 0$

(C) $3a^2 - 26a + 55 = 0$ (D) $3a^2 - 32a + 84 = 0$

Ans. (D)

Solution:

$$\text{Var}(x_i)=\frac{\sum x_i^2}{n}-(\bar{x})^2$$

$$(3.5)^2=\frac{2^2+3^2+a^2+11^2}{4}-\left(\frac{2+3+a+11}{4}\right)^2$$

On solving, we get $3a^2 - 32a + 84 = 0$

Pankers Academy JEE

Illustration 34

The mean and variance of 7 observation are 7 and $\frac{100}{7}$. If 5 of the observations are 2, 4, 7, 11, 10, find the remaining 2 observations.

(A) 3, 6 (B) 3, 12 (C) 4, 11 (D) 5, 10

Ans. (B)

Solution:

$$\frac{\sum x_i}{7}=7$$

$2 + 4 + 7 + 11 + 10 + a + b = 49 \Rightarrow a + b = 15$

$$\frac{\sum(x_i-\bar{x})^2}{7}=\frac{100}{7}$$

$\Rightarrow 25 + 9 + 0 + 16 + 9 + (7 - a)^2 + (7 - b)^2 = 100$

$\Rightarrow (7 - a)^2 + (7 - b)^2 = 41 \Rightarrow a = 3$ and $b = 12$

VARIANCE (FOR FREQUENCY DISTRIBUTION):

Formulae for Variance:

(ii) For frequency distribution:

$$\sigma^2=\frac{\sum f_i(x_i-\bar{x})^2}{N}$$

$$\sigma_x^2=\frac{\sum f_i x_i^2}{N}-\bar{x}^2=\frac{\sum f_i x_i^2}{N}-\left(\frac{\sum f_i x_i}{N}\right)^2$$

For More Join: @IITJEE_Advanced

$$\sigma_d^2 = \frac{\sum f_i d_i^2}{N} - \left(\frac{\sum f_i d_i}{N}\right)^2$$

$$\sigma_u^2 = h^2\left[\frac{\sum f_i u_i^2}{N} - \left(\frac{\sum f_i u_i}{N}\right)^2\right]$$

Note:

(i) $\sigma_x^2 = \sigma_d^2 = h^2\sigma_u^2 = \sigma^2$

(ii) The variance of first n natural numbers is $\boxed{\sigma^2 = \frac{n^2-1}{12}}$

(iii) The variance of A.P. $a, a+d, a+2d, \ldots\ldots\ldots, a+2nd$ is $\boxed{\sigma^2 = \frac{n(n+1)}{3}d^2}$

Properties of Variance:

(i) If the variance of $x_1, x_2, \ldots\ldots, x_n$ is σ^2 then

- var. of $x_1 \pm \lambda, x_2 \pm \lambda, \ldots\ldots\ldots, x_n\lambda$ is σ^2
- var. of $\lambda x_1, \lambda x_2, \ldots\ldots\ldots, \lambda x_n$ is $\lambda^2\sigma^2$
- var. of $ax_1 + b, ax_2 b, \ldots\ldots\ldots, ax_n b$ is $a^2\sigma^2$ (where $\lambda, a, b,$ are constant)

i.e. the variance and $S.D.$ doesn't depend on change of origin (add & subtract) but they are dependent on change of scale (Multiplication & Division).

(ii) Coefficient of $S.D.$ = $\frac{\sigma}{\bar{x}}$

(iii) Coefficient of variation = $\frac{\sigma}{\bar{x}} \times 100$ (in percentage)

Rankers Academy JEE

Illustration 35

$S.D.$ of distribution $41, 42, 43, \ldots. 55$ is

Solution:

Variance $(41, 42, 43, \ldots. 55)$ = Variance $(1, 2, \ldots. 15)$

$= \frac{15^2-1}{12}$

$= \frac{225-1}{12} = \frac{224}{12} = \frac{112}{6} = \frac{56}{3}$

$\therefore\ S.D. = \sqrt{\frac{56}{3}}$

Illustration 36

If $\sum_{i=1}^{11}(x_i - 4) = 11$ and $\sum_{i=1}^{11}(x_i - 4)^2 = 44$, then find variance of $x_1, x_2, x_3 \ldots. x_{11}$.

(A) 4 (B) 3 (C) 7 (D) 11

Ans. (B)

Solution:

$var(x_i - 4) = \frac{44}{11} - \left(\frac{11}{11}\right)^2 = 3$

$\text{var}(x_i) = \text{var}(x_i - 4) = 3$

For More Join: @IITJEE_Advanced

Illustration 37

If $\sum_{i=1}^{18}(x_i-8)=9$ and $\sum_{i=1}^{18}(x_i-8)^2=45$, then find the standard deviation of $x_1, x_2, ...x_{18}$

Solution:

Let $(x_i - 8) = d_i$

$$\therefore \sigma_x = \sigma_d = \sqrt{\frac{\Sigma d_i^2}{n}-\left(\frac{\Sigma d_i}{n}\right)^2} = \sqrt{\frac{45}{18}-\left(\frac{9}{18}\right)^2} = \sqrt{\frac{5}{2}-\frac{1}{4}} = \frac{3}{2}$$

5.5 Mean Square Deviation:

The mean square deviation of a distribution is the mean of the square of deviations of variate from assumed mean. It is denoted by S^2

Hence $S^2 = \frac{\Sigma(x_i-a)^2}{n} = \frac{\Sigma d_i^2}{n}$ (for ungrouped dist.)

$S^2 = \frac{\Sigma f_i(x_i-a)^2}{N} = \frac{\Sigma f_i d_i^2}{N}$ (for freq. dist.) where $d_i = (x_i - a)$

5.6 Relation Between Variance and Mean Square Deviation:

$\because \quad \sigma^2 = \frac{\Sigma f_i d_i^2}{N} - \left(\frac{\Sigma f_i d_i}{N}\right)^2$

$\Rightarrow \quad \sigma^2 = s^2 - d^2$, where $d = \bar{x} - a = \frac{\Sigma f_i d_i}{N}$

$\Rightarrow \quad s^2 = \sigma^2 + d^2 \qquad \Rightarrow s^2 \geq \sigma^2$

Hence, the variance is the minimum value of mean square deviation of a distribution

Bankers' Academy JEE

Illustration 38

Variance of the data given below is

Size of item	3.5	4.5	5.5	6.5	7.5	8.5	9.5
Frequency	3	7	22	60	85	32	8

(A) 1.29 (B) 2.19 (C) 1.32 (D) none of these

Ans. (C)

Solution:

Let the assumed mean be $a = 6.5$

Calculation of variance

x_i	f_i	$d_i = x_i - 6.5$	$f_i d_i$	$f_i d_i^2$
3.5	3	–3	–9	27
4.5	7	–2	–14	28
5.5	22	–1	–22	22
6.5	60	0	0	0
7.5	85	1	85	85
8.5	32	2	64	128
9.5	8	3	24	72
	$N = \Sigma f_i = 217$		$\Sigma f_i d_i = 128$	$\Sigma f_i d_i^2 = 362$

Here $N = 217$, $\Sigma f_i d_i = 128$ and $\Sigma f_i d_i^2 = 362$

$$\therefore \text{ Var } (X) = \left(\frac{1}{N}\Sigma f_i d_i^2\right) - \left(\frac{1}{N}\Sigma f_i d_i\right)^2 = \frac{362}{217} - \left(\frac{128}{217}\right)^2 = 1.668 - 0.347 = 1.321$$

For More Join: @IITJEE_Advanced

Illustration 39

Determine the variance of the following frequency dist.

Class	0–2	2–4	4–6	6–8	8–10	10–12
f_i	2	7	12	19	9	1

Solution:

Let $a = 7, h = 2$

Class	x_i	f_i	$u_i = \frac{x_i - a}{h}$	$f_i u_i$	$f_i u_i^2$
0–2	1	2	–3	–6	18
2–4	3	7	–2	–14	28
4–6	5	12	–1	–12	12
6–8	7	19	0	0	0
8–10	9	9	1	9	9
10–12	11	1	2	2	4
		$N = 50$		$\sum f_i u_i = -21$	$\sum f_i u_i^2 = 71$

$$\therefore \sigma^2 = h^2\left[\frac{\Sigma f_i u_i^2}{N} - \left(\frac{\Sigma f_i u_i}{N}\right)^2\right] = 4\left[\frac{71}{50} - \left(\frac{-21}{50}\right)^2\right] = 4[1.42 - 0.1764] = 4.97$$

Note : Combined variance of two data

- If means of two series containing n_1, n_2 terms are $\bar{x}_1, \bar{x}_2$ and their variance are σ_1^2, σ_2^2 respectively and their combined mean is $\bar{x}$, then the variance σ^2 of their combined series is given by following formula

$$\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{(n_1 + n_2)} \quad \text{where } d_i = \bar{x}_1 - \bar{x}, d_2 = \bar{x}_2 - \bar{x}$$

i.e. $\sigma^2 = \frac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2} + \frac{n_1 n_2}{(n_1 + n_2)^2}(\bar{x}_1 - \bar{x}_2)^2$

Rankers Academy JEE

Illustration 40

The mean of 2 samples of sizes 50 & 40 were found to be 63 and 54. Their variance were 81 & 36. Find the variance of combined sample of size 90

(A) 9 (B) 81 (C) 3 (D) 243

Ans. (B)

Solution:

$\sum x_i = 63 \times 50 = 3150$; $\sum y_i = 40 \times 54 = 2160$

$$\text{var}(x_i) = 81 = \frac{\sum x_i^2}{50} - (63)^2, \quad \text{var}(y_i) = 36 = \frac{\sum y_i^2}{40} - (54)^2$$

$\Rightarrow \sum x_i^2 = 202500$ and $\sum y_i^2 = 118080$

$$\text{Combined variance} = \frac{\left(\sum x_i^2 + \sum y_i^2\right)}{90} - \left(\frac{\sum x_i + \sum y_i}{90}\right)^2$$

$$= \frac{320580}{90} - (59)^2$$

$= 3562 - 3481$

$= 81$

For More Join: @IITJEE_Advanced

Illustration 41

The mean of two groups of sizes 200 and 300 are 25 and 10 respectively. Their standard deviation are 3 and 4 respectively. The variance of combined sample of size 500 is-

(A) 64 (B) 65.2 (C) 67.2 (D) 64.2

Ans. (C)

Solution:

Combined mean $\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2} = \frac{200\times 25 + 300\times 10}{500} = 16$

Here, $d_1 = \bar{x}_1 - \bar{x} = 25 - 16 = 9$ and $d_2 = \bar{x}_2 - \bar{x} = 10 - 16 = -6$

We know that

$$\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2} = \frac{200(9+81) + 300(16+36)}{500} = \frac{33600}{500} = 67.2$$

Illustration 42

The mean and variance of a series containing 5 terms are 8 and 24 respectively. The mean and variance of another series containing 3 terms are also 8 and 24 respectively. The variance of their combined series will be-

(A) 20 (B) 24 (C) 25 (D) 42

Ans. (B)

Solution:

Using $\sigma^2 = \frac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2} + \frac{n_1 n_2}{(n_1 + n_2)^2}(\bar{x}_1 - \bar{x}_2)^2$

$$\Rightarrow \sigma^2 = \frac{5(24) + 3(24)}{5+3} + \frac{5(3)}{(5+3)^2}(8-8)^2 = 24$$

Rankers Academy JEE

5.7 Analysis of Frequency Distributions:

In earlier sections, we have studied about some types of measures of dispersion. The mean deviation and the standard deviation have the same units in which the data are given. Whenever we want to compare the variability of two series with same mean, which are measured in different units, we do not merely calculate the measures of dispersion but we require such measures which are independent of the units. The measure of variability which is independent of units is called coefficient of variation (denoted as C.V.)

The coefficient of variation is defined as

$$C.V. = \frac{\sigma}{x}\times 100, \bar{x} \neq 0,$$

where σ and $\bar{x}$ are the standard deviation and mean of the data.

For comparing the variability or dispersion of two series, we calculate the coefficient of variation for each series. The series having greater C.V. is said to be more variable than the other. The series having lesser C.V. is said to be more consistent than the other.

Comparison of two frequency distributions with same mean Let $\bar{x}_1$ and σ_1 be the mean and standard deviation of the first distribution, and $\bar{x}_2$ and σ_2 be the mean and standard deviation of the second distribution.

For More Join: @IITJEE_Advanced

Then C.V. (1st distribution) = $\frac{\sigma_1}{\bar{x}_1}\times 100$

and C.V. (2nd distribution) = $\frac{\sigma_2}{\bar{x}_2}\times 100$

Given $\bar{x}_1 = \bar{x}_2 = \bar{x}$ (say)

Therefore C.V. (1st distribution) = $\frac{\sigma_1}{\bar{x}}\times 100$ ...(1)

and C.V. (2nd distribution) = $\frac{\sigma_2}{\bar{x}}\times 100$ ...(2)

It is clear from (1) and (2) that the two C.Vs. can be compared on the basis of values of σ_1 and σ_2 only. Thus, we say that for two series with equal means, the series with greater standard deviation (or variance) is called more variable or dispersed than the other Also the series with lesser value of standard deviation (or variance) is said to be more consistent than the other

Note : Coefficient of S.D. = $\frac{\sigma}{\bar{x}}$

Coefficient of variation = $\frac{\sigma}{\bar{x}}\times 100$ (in percentage)

Illustration 43

Find the coefficient of variation of first n natural numbers

Solution:

For first n natural numbers.

Mean ($\bar{x}$) = $\frac{n+1}{2}$, S.D.(σ) = $\sqrt{\frac{n^2-1}{12}}$

Rankers Academy JEE

$\therefore$ coefficient of variance = $\frac{\sigma}{\bar{x}}\times 100 = \sqrt{\frac{n^2-1}{12}}\times\frac{1}{\left(\frac{n+1}{2}\right)}\times 100 = \sqrt{\frac{(n-1)}{3(n+1)}}\times 100$

Illustration 44

Two plants A and B of a factory show following results about the number of workers and the wages paid to them.

	A	***B***
No. of workers	5000	6000
Average monthly wages	Rs 2500	Rs 2500
Variance of distribution of wages	81	100

In which plant, A or B there is greater variability in individual wages ?

Solution:

The variance of the distribution of wages in plant $A(\sigma_1^2) = 81$

Therefore, standard deviation of the distribution of wages in plant $A(\sigma_1) = 9$

Also, the variance of the distribution of wages in plant $B(\sigma_2^2) = 100$

Therefore, standard deviation of the distribution of wages in plant $B(\sigma_2) = 10$

Since the average monthly wages in both the plants is same, i.e., Rs. 2500, therefore, the plant with greater standard deviation will have more variability.

Thus, the plant B has greater variability in the individual wages.

For More Join: @IITJEE_Advanced

Illustration 45

Coefficient of variation of two distributions are 60 and 70 and their standard deviations are 21 and 16, respectively. What are their arithmetic means.

Solution:

Given: C.V. (1st distribution) = 60, $\sigma_1 = 21$

C.V. (2nd distribution) = 70, $\sigma_2 = 16$

Let $\bar{x}_1$ and $\bar{x}_2$ be the means of 1st and 2nd distribution, respectively. Then,

C.V. (1st distribution) = $\frac{c_1}{\bar{x}_1} \times 100$

Therefore, $60 = \frac{21}{\bar{x}_1} \times 100$ or $\bar{x}_1 = \frac{21}{60} \times 100 = 35$

and $C.V.\left(2^{nd} \text{ distribution}\right) = \frac{c_2}{\bar{x}_2} \times 100$

i.e. $70 = \frac{16}{\bar{x}_2} \times 100$ or $\bar{x}_2 = \frac{16}{70} \times 100 = 22.85$

POPOVICIU'S INEQUALITY:

$m < x_i < M$ for all i where $i = 0, 1, 2 \ldots n$

$\bar{x} = \frac{\sum x_i}{n} \Rightarrow \bar{x}$ will also lie between m and M

$(x_i - \bar{x})$ must be bounded

- $|x_i - \bar{x}|$ is maximum at one of the extremes (i.e. m or M)

$\Rightarrow$ either $\bar{x} - m$ or $M - \bar{x}$

on average, distribution of mean to both extremes $= \frac{M-m}{2}$

$\Rightarrow$ furthest any x_i can go from mean is $\frac{M-m}{2}$

$$|x_i - \bar{x}| \le \frac{M-m}{2}$$

$$\Rightarrow \left(x_i - \bar{x}\right)^2 \le \frac{(M-m)^2}{4}$$

$$\Rightarrow \sum_{i=0}^{n} \left(x_i - \bar{x}\right)^2 \le \frac{n(M-m)^2}{4}$$

$$\Rightarrow \sigma^2 \le \frac{(M-m)^2}{4}$$

$$\Rightarrow \text{Variance} \le \frac{(M-m)^2}{4}$$

Rankers Academy JEE

For More Join: @IITJEE_Advanced